



HK IMO Training 2019-2020
Problem Set 1
For 28 September 2019



Problem 1. Let n be a positive integer such that 24 divides $n + 1$. Prove that the sum of the positive divisors of n is divisible by 24.

Problem 2. Let $P(x)$ be a polynomial with integer coefficients. Prove that if there exist integers a, b with $|P(a)| = |P(b)| = 1$ and $|a - b| \geq 3$, then $P(x)$ has no integer roots.

Problem 3. Point D is taken inside $\triangle ABC$ with $AC \neq BC$ such that $\angle ADB = 90^\circ + \frac{1}{2}\angle ACB$. The tangents at point C to the circumcircles of $\triangle ABC$ and $\triangle ADC$ meet the lines AB and AD respectively at points P and Q . Prove that line PQ bisects $\angle BPC$.

Problem 4. In a certain city there are $n \geq 2$ married couples numbered from 1 to n . On a certain day, each man congratulated two women from couples whose numbers differ by one. It turned out that for any two men from couples whose numbers differ by one, there is a woman who received congratulations from both of them. Prove that at least one man congratulated his wife.